

1.1 Areas Between Curves

A Quick Review:

Example: Let A be the area of the region that lies above the x -axis and under the graph of $f(x) = x^2$ between $x = 0$ & $x = 2$. Estimate the area by using a Right Riemann Sum with 4 rectangles.

Definition: If f is a continuous function on $[a, b]$, then the definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

provided that this limit exists and gives the same value for all possible choices of sample points x_i^* . If it does exist, we say that f is **integrable** on $[a, b]$.

Areas Between Two Curves:

General Idea:

Definition (Area Between Two Curves w/respect to x): The area A of the region bounded by the curves $y = f(x)$ & $y = g(x)$, and the lines $x = a$ & $x = b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, is

$$A = \int_a^b [f(x) - g(x)] dx$$

Example 1: Find the area of the region bounded by $y = e^x$, $y = x$, $x = 0$, & $x = 1$.

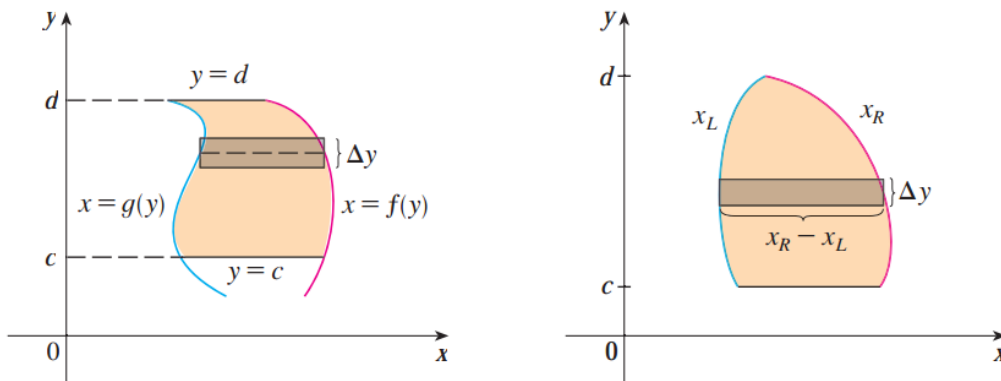
Example 2: Find the area of the region bounded by the curves $y = \sin(x)$, $y = \cos(x)$, $x = 0$, & $x = \frac{\pi}{2}$.

Example 3: Find the area enclosed by the curves $y = x - 1$ & $y^2 = 2x + 6$.

Integrating with Respect to y (instead of x)

General Idea:

We should note that not all curves can be regarded y as a function of x . Some curves are best treated as $x = f(y)$, where x is a function of y . If we consider the picture below we can see that we can still use the idea of a Riemann sum.



Definition (Area Between Two Curves with respect to y): The area A of the region bounded by the curves $x = f(y)$ & $x = g(y)$, and the lines $y = a$ & $y = b$, where f and g are continuous and $f(y) \geq g(y)$ for all y in $[a, b]$, is

$$A = \int_a^b [f(y) - g(y)] dy$$

Example 3 (continued): Find the area enclosed by the curves $y = x - 1$ & $y^2 = 2x + 6$.

Example 4: Find the area enclosed by the curves $y = \sqrt[4]{x}$ & $y = \sqrt{2-x}$ and the x -axis.

a) Find this area by integrating with respect to y .

b) Find this area by integrating with respect to x .